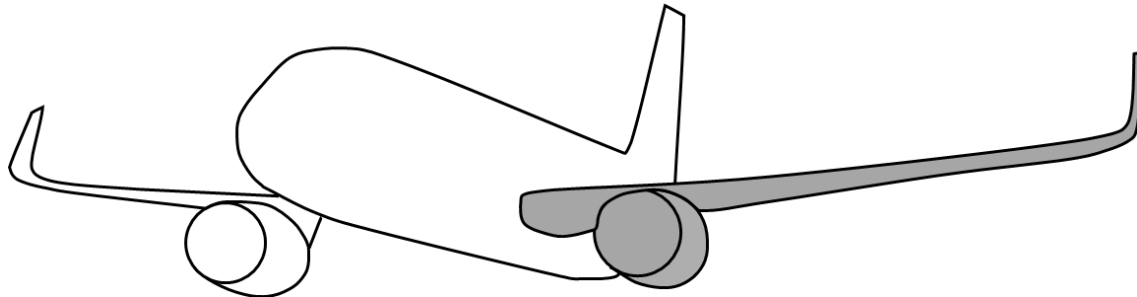


2nd Computational Project



A typical aircraft wing is subjected to many physical phenomena and loads during operation. These include some aerodynamic loads, heat transfer, elastic deformation in the form of bending and torsion. In this project, these specific applications (see **Annex I**) will be analysed by the students, using appropriate simplifications. Each group will be focused on one problem, as presented in the table of **Annex II**. A brief description of each problem is given below.

Heat Conduction Problems:

The objective is to analyse the heat conduction in a flat wing (Problems 1 and 2). The wing is idealized as an isotropic material with known thermal conductivity k . Assume that there is no heat transfer in directions perpendicular to the plane of the wing and that there is convection with the surrounding air with a film coefficient β . The goal is to determine the steady-state temperature distribution in the wing model, as well as the heat flux. The boundary conditions as well as the distribution of sources (f) specified in the statement should be respected.

Fluid Problems:

Consider the potential flow problem (Problem 3), subject to the imposed velocity at the inlet and other specified boundary conditions. The objective is to determine the steady-state velocity potential distribution of the flow, the velocity field, the distribution of pressure isolines throughout the domain (assuming that the inlet pressure is known). In addition, the maximum and minimum values of velocity and pressure in the flow, along with their corresponding locations, are to be determined. Use NACA airfoil data from <http://airfoiltools.com/airfoil/naca4digit>

Torsion Problems:

Consider the torsion of a straight wing, with cross section shown in Problem 4 (NACA airfoil data should be obtained from

<http://airfoiltools.com/airfoil/naca4digit>). The objective is to use the Prandtl stress function to obtain the torsional stiffness constant J , the distribution of shear stresses in the section, shear stress isolines, and to identify the locations and values of the maximum and minimum shear stresses due to a unit torsional moment applied to the wing.

2D Elasticity Problems:

Consider the planar structural wing models of Problems 5 and 6. The aim is to determine the distribution of the displacement field, the strain field, the stress fields in the wing, the various components of the stress tensor, and the principal stresses.

Consider the following properties for the isotropic and homogeneous materials, given in the table:

	E [Pa]	ν	k [W/m°C]	β [W/m ² °C]	ρ_1 [kg/m ³]	ρ_2 [kg/m ³]
2D elasticity/Torsion	20×10^9	0.3			600	2500
Heat transfer			0.6	40		
Fluid potential flow					1.2	

Guidelines for preparation of the work:

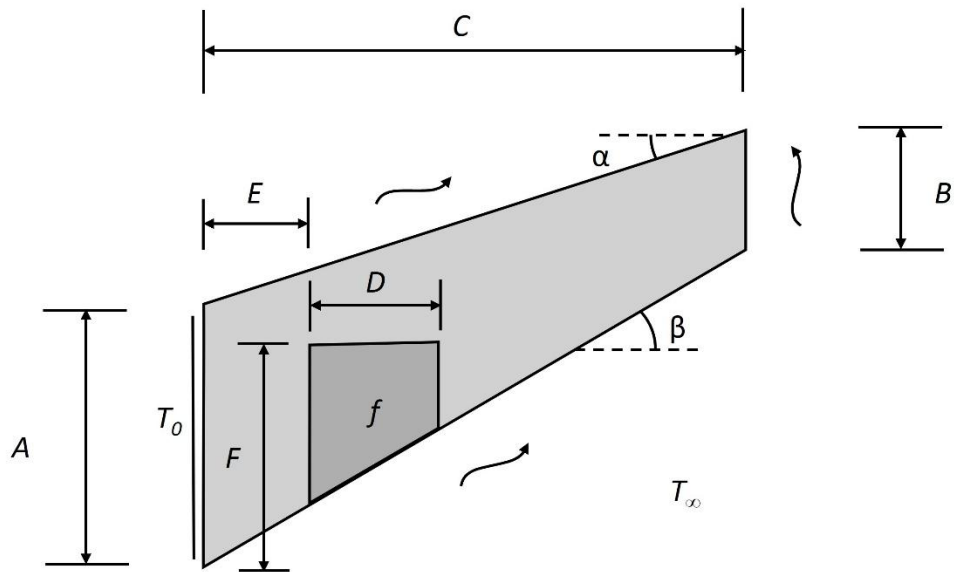
- Present the differential equation that governs the physical phenomenon you want to study and clearly indicate the boundary conditions used and all assumptions you made. You must also clearly indicate in the report all the data of the problem.
- Start by establishing and solving a simple problem in the commercial program ANSYS (APDL interface). The simple problem is chosen by the students and must have a simpler domain (geometry) with the same type of material properties, the same type of boundary conditions, and have an analytical solution, to allow testing the procedure implemented in the commercial program.
- Solve the simple problem in the commercial finite element program ANSYS Student (APDL interface), which can be obtained from <https://www.ansys.com/academic/students>.
- For scalar variable problems (if applicable), use the **PLANE55** elements (T3 and Q4) or **PLANE77** elements (T6 and Q8), according to the problem data. These elements solve 2D thermal problems. To solve fluid and torsion problems, the thermal analogy should be used.
- For 2D elasticity problems (if applicable), use the **PLANE182** elements (T3 and Q4) or **PLANE183** elements (T6 and Q8 elements), according to the problem data.

- After testing the procedure with the simple problem, solve the problem given in the statement using ANSYS APDL and perform a convergence analysis of the solution.
- Based on the provided MATLAB code (**FE2D_GroupNumber.mlx**), implement the routines for calculating the coefficient matrix and second-member vector for your problem and test it with the simple problem. Compare it with the corresponding results (from the simple problem) obtained by ANSYS. An example code using the Q9 element is provided for reference (**Q9.mlx**)
- After testing and validating the results of the MATLAB code for the simple problem, export the data relating to the coordinates of the mesh nodes and their respective nodal connectivity from ANSYS (referring to your problem) to a file to be subsequently read by the MATLAB code (instructions included in the code).
- Obtain the requested results for your problem using the MATLAB code and compare them with the results previously obtained by ANSYS APDL.
- Present the results, discuss them, and present final conclusions in a **report of no more than 10 pages, which must include all the problem data**. Also **submit the ANSYS APDL data file(s) (input), as well as the MATLAB code** used to solve the problem (**replace “GroupNumber” in the name of the MATLAB file with your group number**). The APDL files should, contain commands independent of the selection operations performed during the ANSYS interactive session. You must ensure that the files can be executed without errors (this requirement also applies to the MATLAB code).

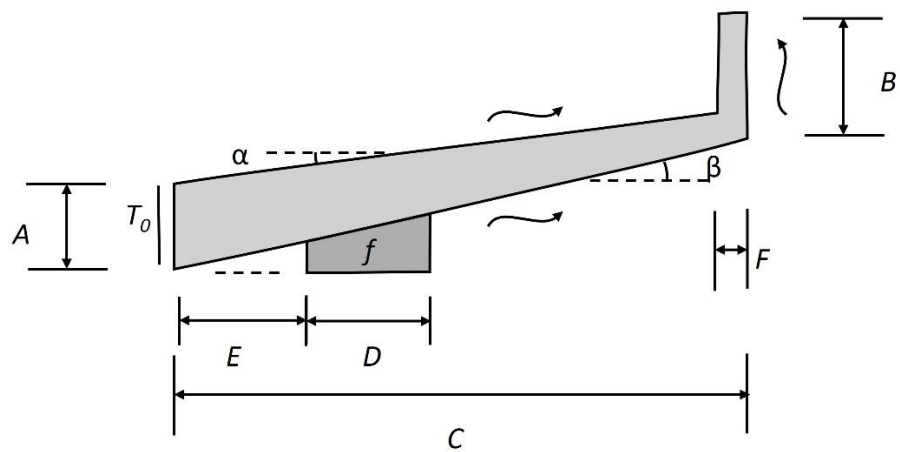
The report must be submitted in PDF format, together with the ANSYS APDL input files and the MATLAB program, combined into a single compressed file (.zip) and uploaded to the Fenix system by January 16, 2026 at 11:59 PM.

ANNEX I

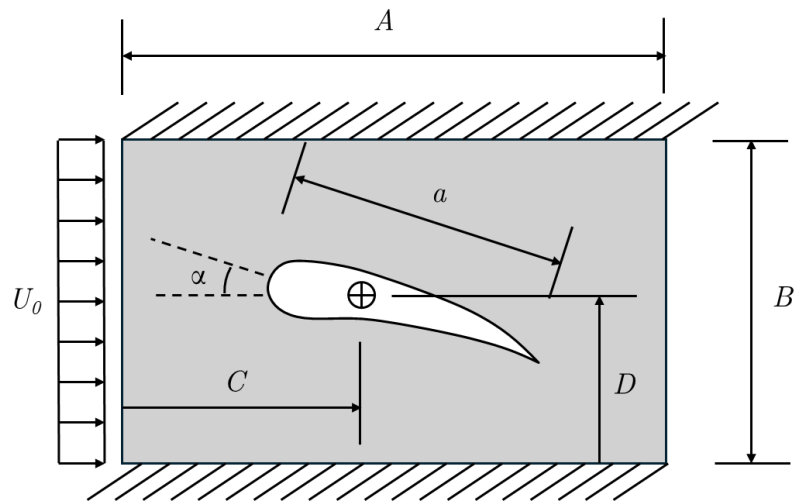
Problem 1 – Heat transfer



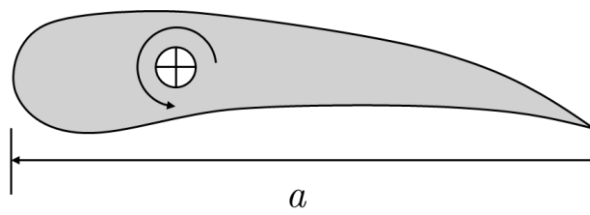
Problem 2 – Heat transfer



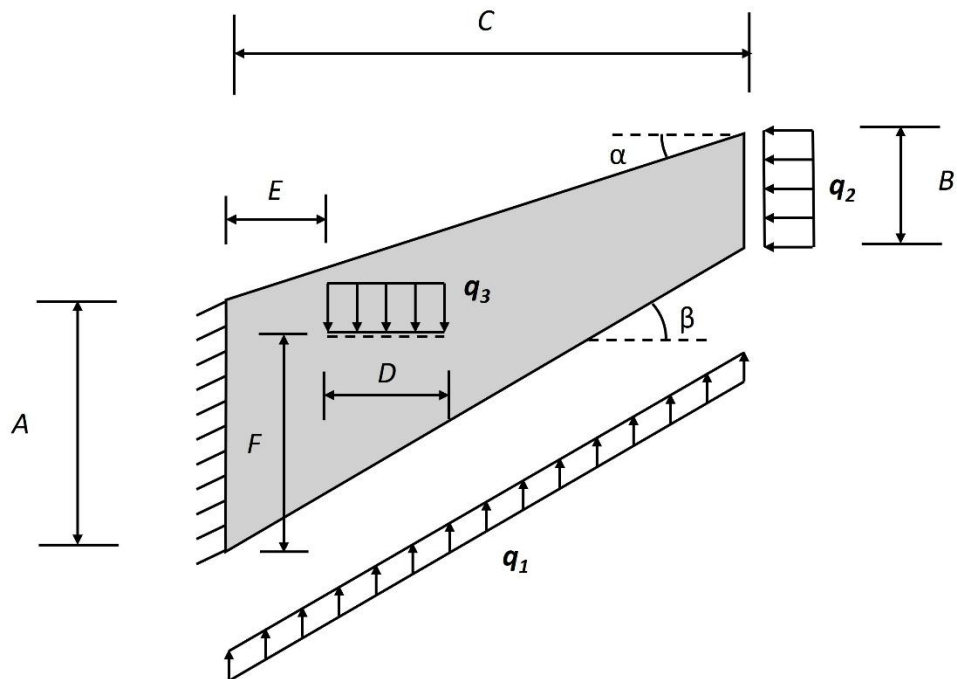
Problem 3 – Fluid mechanics



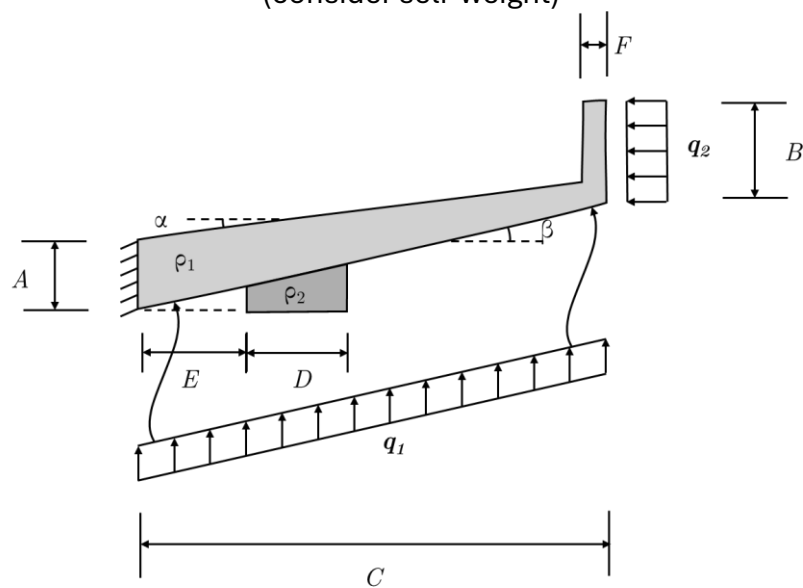
Problem 4 – Torsion



Problem 5 – Elasticity (Plane stress)
(consider 1g take-off acceleration)



Problem 6 – Elasticity (Plane strain)
(consider self-weight)



ANNEX II

Group	Element	Problem	A [m]	B [m]	C [m]	D [m]	E [m]	F [m]	t [m]	α [°]	β [°]	a [m]	U_0 [m/s]	f [kW/m ²]	T_0 [°C]	T_∞ [°C]	q_1 [Pa]	q_2 [Pa]	q_3 [Pa]	NACA
1	T3	1	1.25	0.625	5	0.6	2	1	0.01		15			3	20	5				
2	T6	2	0.2	0.6	5	0.6	2	0.1	1	14.5	15			4	20	5				
3	Q4	3	10	5	5	2.5				0		0.5	20							0006
4	Q8	4										0.5								0006
5	T3	5	1.25	0.625	5	0.7	2.2	0.9	0.01		16						1.0E+04	5.0E+03	1.5E+06	
6	T6	6	0.2	0.6	5	0.7	2.2	0.1	1	15.5	16						1.0E+03	5.0E+02		
7	Q4	1	1.5	0.75	6	0.8	3	1.1	0.01		17			5	20	0				
8	Q8	2	0.2	0.6	6	0.8	3	0.1	1	16.55	17			6	20	0				
9	T3	3	12	6	6	3				5		0.6	15							0008
10	T6	4										0.6								0008
11	Q4	5	1.5	0.75	6	0.75	2.8	1.2	0.01		18						1.2E+04	6.0E+03	1.8E+06	
12	Q8	6	0.2	0.6	6	0.8	3	0.1	1	16.55	17						1.2E+03	6.0E+02		
13	T3	1	1.75	0.875	7	0.9	3	1.5	0.015		19			4	20	-5				
14	T6	2	0.15	0.5	7	0.9	3	0.15	1	18.6	19			6	20	-5				
15	Q4	3	14	7	7	3.5				6		0.7	17							0009
16	Q8	4										0.7								0009
17	T3	5	1.75	0.875	7	0.85	3.5	1.45	0.015		20						1.4E+04	7.0E+03	2.1E+06	
18	T6	6	0.15	0.5	7	0.9	3	0.15	1	18.6	19						1.4E+03	7.0E+02		
19	Q4	1	2	1	8	1	4	1.8	0.015		21			3	20	-10				
20	Q8	2	0.3	0.9	8	1	4	0.15	1	20.65	21			6	20	-10				
21	Q8	3	14	7	6	4				7		0.8	22							0010
22	T3	4										0.8								0010
23	T6	5	2	1	8	1.1	3.5	1.5	0.015		22						1.6E+04	8.0E+03	2.4E+06	
24	Q4	6	0.3	0.9	8	1	4	0.15	1	20.65	21						1.6E+03	8.0E+02		
25	Q8	1	2.25	1.125	9	1.1	4.5	2	0.02		23			4	20	-20				
26	T6	2	0.4	1.2	9	1.1	4.5	0.2	1	22.7	23			5	20	-20				
27	T3	3	14	7	6	4				0		0.8	22							0012
28	T6	4										0.8								0012
29	Q4	5	2.25	1.125	9	1	4.4	1.9	0.02		24						1.8E+04	9.0E+03	2.7E+06	
30	Q8	6	0.4	1.2	9	1.1	4.5	0.2	1	22.7	23						1.8E+03	9.0E+02		
31	Q4	1	2.5	1.25	10	1.2	5	2.2	0.02		25			6	10	5				
32	T3	2	0.4	1.2	10	1.2	5	0.2	1	24.75	25			5	10	5				
33	T6	3	10	5	6	4				4		0.6	20							0015
34	Q4	4										0.6								0015
35	Q8	5	2.5	1.25	10	1.25	5	2.3	0.02		26						2.0E+04	1.0E+04	3.0E+06	
36	Q4	6	0.4	1.2	10	1.2	5	0.2	1	24.75	25						2.0E+03	1.0E+03		
37	T3	1	2.25	1.125	9	1.2	4	1.9	0.02		27			1	10	0				
38	T6	2	0.4	1.2	9	1.2	4	0.2	1	26.7	27			2	10	0				
39	Q4	3	10	5	6	4				6		1.2	20							0018
40	Q8	4										1.2								0018
41	T3	5	2.25	1.125	9	1.25	4.5	2	0.02		28						1.8E+04	9.0E+03	2.7E+06	
42	T3	6	0.4	1.2	9	1.2	4	0.2	1	26.7	27						1.8E+03	9.0E+02		
43	T6	1	2	1	8	0.9	4	1.8	0.015		29			3	10	-5				
44	Q4	2	0.3	0.9	8	0.9	4	0.15	1	28.65	29			4	10	-5				

ANNEX II

Group	Element	Problem	A [m]	B [m]	C [m]	D [m]	E [m]	F [m]	t [m]	α [°]	β [°]	a [m]	U_0 [m/s]	f [kW/m ²]	T_0 [°C]	T_∞ [°C]	q_1 [Pa]	q_2 [Pa]	q_3 [Pa]	NACA
45	Q8	3	12	6	5	2				0		1.3	15							0021
46	Q8	4										1.3								0021
47	T3	5	2	1	8	0.95	3.5	1.75	0.015		30						1.6E+04	8.0E+03	2.4E+06	
48	T6	6	0.3	0.9	8	0.9	4	0.15	1	28.65	29						1.6E+03	8.0E+02		
49	Q4	1	1.75	0.875	7	0.9	3	1.5	0.015		31			1	10	-10				
50	Q8	2	0.3	0.9	7	0.9	3	0.15	1	30.6	31			6	10	-10				
51	Q8	3	10	6	5	4				5		0.7	15							0024
52	T3	4										0.7								0024
53	T6	5	1.75	0.875	7	0.87	3	1.4	0.015		32						1.4E+04	7.0E+03	2.1E+06	
54	Q4	6	0.3	0.9	7	0.9	3	0.15	1	30.6	31						1.4E+03	7.0E+02		
55	Q8	1	1.5	0.75	6	0.8	3	1.3	0.01		33			6	15	5				
56	T6	2	0.2	0.6	6	0.8	3	0.1	1	32.55	33			1	15	5				
57	T3	3	8	6	4	3				2		0.75	22							1408
58	T6	4										0.75								1408
59	Q4	5	1.5	0.75	6	0.7	2.5	1.2	0.01		34						1.2E+04	6.0E+03	1.8E+06	
60	Q8	6	0.2	0.6	6	0.8	3	0.1	1	32.55	33						1.2E+03	6.0E+02		
61	Q4	1	1.25	0.625	5	0.7	2.5	1.1	0.01		35			3	15	0				
62	T3	2	0.2	0.6	5	0.7	2.5	0.1	1	34.5	35			5	15	0				
63	T6	3	9	7	4	3				1		0.65	19							1410
64	Q4	4										0.65								1410
65	Q8	5	1.25	0.625	5	0.6	2.4	1	0.01		36						1.0E+04	5.0E+03	1.5E+06	
66	T3	6	0.2	0.6	5	0.7	2.5	0.1	1	34.5	35						1.0E+03	5.0E+02		
67	T3	1	1.5	0.75	6	0.7	3	1.3	0.01		37			2	15	-5				
68	T6	2	0.2	0.6	6	0.7	3	0.1	1	36.55	37			5	15	-5				
69	Q4	3	8	5	3	3				5		0.8	19							1412
70	Q8	4										0.8								1412
71	Q4	5	1.5	0.75	6	0.8	2.5	1.2	0.01		38						1.2E+04	6.0E+03	1.8E+06	
72	T3	6	0.2	0.6	6	0.7	3	0.1	1	36.55	37						1.2E+03	6.0E+02		
73	T6	1	1.75	0.875	7	0.8	3	1.5	0.015		39			4	15	-10				
74	Q4	2	0.3	0.9	7	0.8	3	0.15	1	38.6	39			2	15	-10				
75	Q8	3	10	6	3	3				0		1	19							2408
76	Q8	4										1								2408
77	T3	5	1.75	0.875	7	0.85	3.5	1.45	0.015		40						1.4E+04	7.0E+03	2.1E+06	
78	T6	6	0.3	0.9	7	0.8	3	0.15	1	38.6	39						1.4E+03	7.0E+02		
79	Q4	1	2	1	8	0.9	4	1.8	0.015		39			2	15	-20				
80	Q8	2	0.3	0.9	8	0.9	4	0.3	1	38.65	39			1	15	-20				
81	T3	3	9	6	3	3				2		0.9	20							2410
82	T6	4										0.9								2410
83	Q4	5	2	1	8	1	3.8	1.7	0.015		38						1.6E+04	8.0E+03	2.4E+06	
84	Q8	6	0.3	0.9	8	0.9	4	0.3	1	38.65	39						1.6E+03	8.0E+02		
85	T3	1	2.25	1.125	9	1.1	4.5	2	0.015		37			2	15	-25				
86	T6	2	0.3	0.9	9	1.1	4.5	0.15	1	36.7	37			2	15	-25				
87	Q4	3	9	6	4	3				5		0.9	20							2411
88	Q8	4										0.9								2411

ANNEX II

Group	Element	Problem	A [m]	B [m]	C [m]	D [m]	E [m]	F [m]	t [m]	α [°]	β [°]	a [m]	U_0 [m/s]	f [kW/m ²]	T_0 [°C]	T_∞ [°C]	q_1 [Pa]	q_2 [Pa]	q_3 [Pa]	NACA
89	T3	5	2.25	1.125	9	1	4.4	1.9	0.02		36						1.8E+04	9.0E+03	2.7E+06	
90	T6	6	0.3	0.9	9	1.1	4.5	0.15	1	36.7	37						1.8E+03	9.0E+02		
91	Q4	1	2.5	1.25	10	1.2	5	2.1	0.02		35			3	25	-20				
92	Q8	2	0.4	1.2	10	1.5	5	0.2	1	34.75	35			3	25	-20				
93	T3	3	11	7	4	3				3		0.9	20							2412
94	T6	4										0.9								2412
95	Q4	5	2.5	1.25	10	1.25	4.5	2	0.02		34						2.0E+04	1.0E+04	3.0E+06	
96	Q8	6	0.4	1.2	10	1.5	5	0.2	1	34.75	35						2.0E+03	1.0E+03		
97	T6	1	2.25	1.125	9	1.2	4	1.5	0.02		33			4	25	-10				
98	T3	2	0.4	1.2	9	1.2	4	0.2	1	32.7	33			4	25	-10				
99	T6	3	11	7	4	3				3		0.8	20							2414
100	Q4	4										0.8								2414